

Mayank Chandak

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 [GitHub](#)

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EDUCATION

Degree	Institute	Board / University	CGPA/%	Year
B.Tech Mechanical Engineering	Indian Institute of Technology Madras	IITM	9.43*/10	2026 (Expected)
Senior Secondary	Khar Education Society, Mumbai	HSC	95.0% (rank 3)	2022
Matriculation	Arya Vidya Mandir Juhu, Mumbai	ICSE	98.33% (rank 2)	2020

POSTERS AND PUBLICATIONS

- [1] **M. Chandak**, et al., "*FlareGS: 4D Flare Removal using Gaussian Splatting for Urban Scenes*", Accepted at ICCV 2025 Workshop on Challenge Of Out-Of-Label Hazards in Autonomous Driving (2COOOL)
- [2] **M. Chandak**, et al., "*FlareGS: 4D Flare Removal using Gaussian Splatting for Urban Scenes*", Poster presented at ICCV 2025 Workshop on Challenge Of Out-Of-Label Hazards in Autonomous Driving (2COOOL)
- [3] **M. Chandak**, "*Flare Removal from Videos*", Poster presented at Young Research Fellow Poster Exhibition, IIT Madras, India, 2025

EXPERIENCE

Undergraduate Researcher || AUTONOMOUS CAMERA CONTROL FOR LAPAROSCOPIC SURGERY **May 2025 – Present**
PI: Dr. Nirav Patel INSPIRE Lab, IIT Madras, India

- Developing a **reinforcement learning pipeline** for **autonomous control** of a **pivotized laparoscopic camera** in a multi-agent surgical simulation, coordinating with two tool manipulators performing subtasks like gallbladder lifting and cauterization.
- Developed a **pivotized TPSD camera module** with Remote Center of Motion (RCM) constraints in a **SOFA-based environment**.
- Modeled the camera as an **independent agent** with action space, perception, and tissue-aware constraints for adaptive viewpoints.
- Designing a **reward strategy** to optimize surgical site visibility, minimize occlusions, and align with tool trajectories.

Young Research Fellow || FLARE REMOVAL, GAUSSIAN SPLATTING, TEMPORAL VIDEO RESTORATION **Dec 2024 – Mar 2025**
FlareGS: 4D Flare Removal using Gaussian Splatting for Urban Scenes Computational Imaging Lab, IIT Madras, India
PI: Dr. Kaushik Mitra

- Developed a **physics-based synthetic flare generation pipeline** modeling ghost flares using **Fraunhofer diffraction** and **Mie scattering** theory, enabling robust training of perception systems without paired datasets.
- Implemented a **depth-aware UFormer** restoration module using **RGB + LiDAR input** to jointly predict flare and clean components, leveraging **time-of-flight depth** as an optical-invariant prior for precise flare separation.
- Integrated **4D Gaussian Splatting** for **temporally consistent scene reconstruction** from multi-camera input, enabling **novel view synthesis** critical for path planning and obstacle avoidance in autonomous robotic systems.
- Conducted evaluation on a **1,000-frame benchmark** combining **real-world flares** (Flare7K++) and synthetic corruptions on **Waymo urban driving videos**, stratified across **M1–M3 intensity levels**.

VidFlareNet: A Deep Learning Framework for Video Flare Removal **Jul 2024 – Sep 2024**
PI: Dr. Kaushik Mitra Computational Imaging Lab, IIT Madras, India

- Developed **VidFlareNet**, a video restoration framework integrating **spatial transformers**, **Mamba temporal modules**, and **depth-guided attention** to remove **flare artifacts** while preserving fidelity and temporal consistency.
- Incorporated **depth priors** via a **depth-reweighting attention module** to recover flare-occluded regions.
- Built a **2,600-video flare dataset** by injecting dynamic **Flare-R** artifacts into **YouTubeVOS** for large-scale training.
- Outperformed **state-of-the-art models** (Restormer, Uformer) with higher **PSNR (20.03 vs 17.89)**, lower **Short-Term Loss (8.24 vs 9.12)**, and improved downstream **detection and optical flow** on synthetic dataset.

Undergraduate Researcher || LOW LIGHT IMAGE ENHANCEMENT **Feb 2024 – May 2024**
PI: Dr. Kaushik Mitra Computational Imaging Lab, IIT Madras, India

- Built **sub-90K parameter Retinex and U-Net models** in **PyTorch** for extreme low-light enhancement, enabling **real-time inference**.
- Contributed to the **NTIRE HDR** and **PBDL low-light competitions**, exploring **transformer and diffusion models** for high-resolution (**6K×4K**) enhancement, and achieved a **top-10 global rank** in the PBDL sRGB leaderboard.
- Reproduced and optimized pipelines from **10+ state-of-the-art methods** (e.g., SID, IAT, Retinexformer), tailoring them for custom datasets and benchmarks.

MLE Intern || GENAI & OCR FOR DOCUMENT PARSING **Jun 2024 – Jul 2024**
Quantiphi Analytics Solutions Pvt. Ltd. Mumbai, India

- Trained in **Python, LLMs, and GenAI**, developing **customized client-facing AI solutions** and **optimized model performance**.
- Researched advanced **OCR techniques** for **tabular data extraction**, conducting in-depth analysis of **Sparrow Parse**.
- Collaborated with cross-functional teams to integrate **AI strategies** into production pipelines, improving scalability and efficiency.

NOTABLE PROJECTS

MyoFootball: Reward-Driven Musculoskeletal Control for Goal-Oriented Locomotion (Watch here)

Independent Project Inspired by Prof Vikash Kumar, CMU

- Developed **MyoFootball**, a custom extension of **MyoSuite’s musculoskeletal MyoLeg model**, integrating **ball dynamics, contact mechanics, and goal-scoring tasks** to simulate end-to-end sports-like locomotion and kicking behaviors.
- Designed a **multi-stage reward pipeline** that jointly optimized **gait stability, ball approach, impact precision, and post-kick balance**, carefully mitigating failure modes such as leg locking, dragging, and reward exploitation.
- Trained agents to achieve **80% goal-scoring accuracy**, with emergent **coordinated locomotion and targeted kicking behaviors** demonstrated through **100+ reward design iterations**, validating robustness across randomized ball placements.
- Executed independently in under 4 weeks while managing research and coursework, gaining deep expertise in **MyoSuite environment design, musculoskeletal RL training**, and handling **reward shaping pitfalls** in sports-inspired motor control.

TD3 Reinforcement Learning for Robotic Arm Task in Robosuite

- Developed a **TD3-based reinforcement learning agent** to control the **“Panda” robotic arm** in the **Robosuite environment**, enabling it to perform **complex manipulation tasks** such as opening a door with **high precision and efficiency**.
- **Constructed the TD3 architecture** from the **ground up** using **PyTorch**, including **actor and critic networks, target networks**, and an **experience replay buffer**, resulting in a fully functional RL agent capable of handling **robotic control tasks**.
- Conducted extensive **hyperparameter tuning** through **multiple experiments**, refining **learning rates, batch sizes, layer sizes**, and **discount factors** based on **theoretical insights**, significantly improving the agent’s **training stability and performance**.

“Ant-Based Swarming with Positionless Micro Air Vehicles for Communication Relay” Reproduction and Extension

Course Project – ME5253: Network Dynamics

- Reproduced the original **ant-inspired swarm algorithm** where positionless MAVs self-organize into a **communication relay network** through **virtual pheromone, probabilistic exploration, and local neighbor coordination** inspired by army ant foraging.
- Implemented and validated the **pheromone update, hop-based routing, and evaporation-driven retraction mechanisms** in Python, accurately replicating the paper’s results on **connectivity and coverage** across multiple swarm sizes.
- Proposed an improvement addressing **wind-induced drift** by introducing a **dynamic role-swapping and local correction mechanism**, significantly enhancing network stability and reattachment under simulated wind disturbances.

MADDPG Algorithm Implementation: Multi-Agent Actor-Critic Framework in PyTorch

- Implemented MADDPG from the **“Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments”** paper in **PyTorch**, leveraging the **CTDE paradigm** with independent actors, centralized critics, and a shared multi-agent replay buffer.
- Enhanced training stability and exploration by integrating **target network synchronization and adaptive noise**, enabling robust policy learning in partially observable, cooperative-competitive multi-agent environments.

Multi-Agent Warehouse Automation

Course Project – ME3302: Automation in Manufacturing

- Built a custom **grid-based warehouse environment** with **dynamic shelf activations**, where multiple agents collaboratively perform pickups and deliveries under realistic navigation, manipulation, and operational constraints.
- Developed a **CTDE framework** with a centralized assignment policy and decentralized agents using **local observations**, supported by a shaped **reward function** optimizing task completion, idle time, and collisions to achieve **scalable efficiency gains**.

TECHNICAL SKILLS

Programming Languages: Java, C++, Python, MATLAB, Bash/zsh

ML & AI Frameworks: TensorFlow, PyTorch, PyTorch Lightning, Keras, HuggingFace, LangChain, Streamlit

Data Science & Utilities: Scikit-learn, OpenCV, Pillow, SciPy, Pandas, Numpy, Matplotlib, Seaborn

Simulation, Robotics & CAD: MuJoCo, ROS, CoppeliaSim, SOFA Simulation, AutoCAD, Creo, SolidWorks, Fusion 360

Productivity Software: macOS, Windows, Linux (Ubuntu), Git (GitLab & GitHub), VSCode, Overleaf, Canva, L^AT_EX

NOTABLE ACHIEVEMENTS

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| • Young Research Fellow (YRF) Award (~ ₹100K) , IIT Madras
<i>Selected as one of 17 Young Research Fellows from a pool of 300+ students for funded undergraduate research.</i> | 2024 |
| • Ranked Top 7 out of 199 students in the Mechanical Engineering Department, IIT Madras | 2024 |
| • Ranked Top 5% nationally among 150,000 candidates in JEE Advanced, one of India’s most competitive exams | 2022 |
| • Ranked Top 1% nationally among 1 million candidates in JEE Mains, India’s largest engineering entrance test | 2022 |
| • Grand Finale Qualifier in the Clash of PI National Mathematics Contest, representing school at the national level | 2018 |
| • All India Rank 1 in NFLAT(National Finance Literary Assessment Test) by NCFE
<i>Awarded ₹10,000 cash prize and a laptop for rank 1 in a nationwide financial literacy exam.</i> | 2018 |
| • Outstanding Performance Award , ASSET Exams (Top 1 percentile yearly) | 2013–2019 |

RELEVANT COURSEWORK

Mathematics: Multivariable Calculus, Series & Matrices, Differential Equations, Probability & Statistics, Stochastic Processes

Robotics/Controls: Kinematics & Dynamics, Automation in Manufacturing, Control Systems*, Mechatronics*, Network Dynamics*

AI/ML: Mathematical Foundation of Data Science, Machine Learning, Reinforcement Learning, Data Analytics Lab*

*Ongoing as of October 2025.